Auditable Remaining Useful Life Prediction Under Correlated Sensor Failures: Guarded Recovery for Turbofan Engines

Devanshu Verma
Department of DSAI
IIIT-Naya Raipur
Email: devanshu24102@iiitnr.edu.in

Devesh Gour
Department of DSAI
IIIT-Naya Raipur
Email: devesh24102@iiitnr.edu.in

Vishal Agrawal
Department of DSAI
IIIT-Naya Raipur
Email: vishal24102@iiitnr.edu.in

***Abstract*—Remaining useful life (RUL) prediction must distinguish degradation from measurement failures across operating conditions. GRACE-RUL combines local CNN–GRU features, residual BiLSTM sensor recovery, Transformer context, an independent finite-fault detector and a validation-audited RUL path. Format 10 adds an observed engine-age channel and supervises continuity on the deployed repair path; it disables the bounded corrector after a null component ablation and replaces the hard repair threshold with probability-weighted soft gating. One shared model pools all four NASA C-MAPSS subsets with 567 fitting, 36 selection, 35 calibration and 71 internal-test engines. Faults follow related-channel, temporally persistent mechanisms in raw sensor units. Conservative thresholds, fitting-only support bounds and a complete 41-condition audit limit healthy-sensor displacement, with refined detector recall across drift, bias, noise and stuck faults. Predictor refinement uses active-path paired-prefix continuity, fitting-error tails and corruption-aware RUL supervision. A validation-selected candidate is frozen before interval calibration and test evaluation. On 707 official endpoints its capped RMSE is 10.50 cycles, MAE is 6.96, and severity-0.2 stuck-fault RMSE averages 12.49 over three corruption replicates. Internal clean RMSE is 11.80 over 484 prefixes from 71 engines. Observable interval scales are fitted independently of engine-level conformal calibration. Raw RMSE is 20.47; 150 engines with raw RUL above 125 account for 77.5% of raw squared error. A validation-selected monotone plateau stretch (alpha=0.150 above 105 cycles) reduces raw RMSE from 67.30 to 66.03 on selection engines while keeping capped RMSE within its guard. A previously inspected severe prediction failure is reported explicitly. Under broadband sensor noise at 20% of the within-regime scale, capped RMSE rises from 10.50 to 11.44 cycles with 99.4% empirical coverage; at 50% it is 13.44 cycles. Reused benchmarks, one training seed and simulated faults limit generalization claims.**

***Index Terms*—Remaining useful life, C-MAPSS, correlated sensor faults, guarded recovery, pooled training, CNN–GRU, Transformer, conformal calibration.**

I. Introduction

Turbofan RUL estimation maps observed operating history to remaining cycles before failure. Missing sensor readings are observable, while a finite bias, drift or stuck reading can resemble degradation. Operating-condition changes make this distinction harder. C-MAPSS provides a reproducible simulation benchmark, but added measurement faults are modeling assumptions rather than recorded aircraft incidents [1].

This manuscript presents an auditable prognostic pipeline and an empirical evaluation on reference and structured sensor-fault NASA C-MAPSS telemetry, including observed recovery limitations and an explicit failure analysis.

GRACE-RUL is organized around three design commitments: local convolutional–recurrent modeling of recent degradation, selective recurrent recovery of corrupted readings, and a Transformer context branch. The current format-10 implementation adds an observed engine-age channel and a probability-weighted finite-repair gate to one shared model across FD001–FD004, without requiring a subset label at deployment. Its contributions are a prefix-consistent related-channel fault simulator; independent finite-fault detection with soft, support-bounded repair; quality-conditioned local/global RUL fusion; validation-led predictor refinement; independently calibrated observable intervals; and a broadband measurement-noise and channel-dropout benchmark that reports accuracy and calibrated coverage without refitting. Complete recovery checks and state-bound audits make the selected path inspectable. Severe prediction failures, reused benchmarks and unvalidated fleet assumptions are retained explicitly in the evidence.

II. Related Work

A) Local Degradation Modeling and Interpretation

The supplied hybrid CNN–GRU study emphasizes local feature extraction, temporal degradation modeling, and sensor-level interpretation [2]. GRACE-RUL retains a CNN–GRU branch over the most recent 30 cycles. Interpretation is implemented as optional post-training KernelSHAP/LIME analysis [5], [6]; it is not a term in the neural training objective. That work's complete reliability/generalization/explainability formulation is not reproduced literally.

B) Recovery Under Partial Sensor Failure

The partial-failure study motivates paired corrupted/reference histories, bidirectional recurrent recovery, and a training-only adversarial critic [3]. The implementation adds a deployment-available finite-fault detector and availability-based selective blending, because synthetic altered-entry labels are unavailable at inference. WGAN-GP supplies the

adversarial training objective [7]. Original C-MAPSS readings remain noisy reference measurements, rather than pristine sensor truth.

C) Transformer Context and Residual Compensation

Trans-D3 motivates combining a Transformer representation with prediction correction [4]. The current implementation instead uses a bounded supervised residual MLP only as a retained, zero-bound ablation artifact; after a null component ablation it is disabled in the deployed format-10 path, so the final RUL is the branch-fusion base prediction. The study also uses physical degradation slew and anti-rebound loss penalties on paired engine prefixes. It contains no TD3 actor, twin action-value critics, replay buffer, policy update, or LQR filter. The separate WGAN critic evaluates sensor sequences during recovery training and is unrelated to reinforcement learning.

III. Problem Statement and Data Protocol

A) Observed Histories, Targets and Masks

An observation contains three operating settings and 21 sensors. Engine ID is never a neural feature; the observed cycle index is used only to derive the normalized age channel in format 10. A window ending at t contains at most $L = 70$ already observed cycles, left-padded to fixed length. Validity V masks padding and availability M marks finite sensor readings. Normalized settings are U and encoded observed sensors are S; nonfinite measurements are replaced by zero after the mask is constructed. A zero measurement remains valid. For a complete fitting engine ending at T_e, raw RUL is $r_{e,t} = T_e - t$ and

$$y_{e,t} = \min(125, r_{e,t}). \tag{1}$$

Official RUL labels enter evaluation after freeze. Targets, future observations, source identity and simulated altered-entry labels never enter inference. Softplus estimates may exceed 125: the cap specifies the target, not a hard prediction ceiling or a guarantee about future failure time.

B) One Pooled Engine-Disjoint Split

The available four training files are pooled after namespacing engine IDs for bookkeeping. Split seed 42 assigns approximately 80% to fitting, 10% to a selection/calibration pool and 10% to internal test. Half of the development pool is reserved for independent interval calibration. Exact rounded counts are 567 fitting, 36 selection, 35 calibration and 71 internal-test engines. Every prefix of an engine stays in its partition. Per-FD reporting is an evaluation breakdown, not a collection of different models.

C) Fitting-Only Regimes and Encoding

Operating settings are standardized using fitting engines only. GMM BIC selects the regime count from fitting settings, followed by K-means assignment; the selected shared scaler has 6 regimes. Per-regime sensor means/scales include measurement-resolution floors. For raw sensor x_j in assigned regime k,

$$z_j = (x_j - \mu_{kj})/\sigma_{kj}, \tag{2}$$

$$e_j = s \operatorname{asinh}(z_j/s), \qquad s = 1, \tag{3}$$

$$\phi^{-1}(e_j) = s \sinh(e_j/s). \tag{4}$$

This invertible encoding compresses extreme standardized values; it does not reduce their physical raw-unit error. Faults are applied after decoding and only changed cells are re-encoded. Intact encoded cells retain their bits. Recovery support is the empirical per-regime/channel fitting range, not a certified manufacturer operating envelope. Selection, calibration and test engines fit none of these statistics.

D) Causal Prefix Evaluation

Fitting uses engine-balanced sampling from eligible within-engine windows. Selection and internal testing use a fixed inspection grid, beginning at cycle 30 with stride 30, independent of eventual failure. Calibration evaluates every observed prefix of each reserved complete engine. Official test evaluation uses its final observed endpoint. Bidirectional recovery and non-triangular attention operate inside the already observed window; unseen future readings are absent. The local branch uses 30 cycles and the global branch 70. Neither enforces monotonic predictions at inference.

E) Data Integrity and Evidence Boundaries

The shared model receives only normalized operating settings, observed sensors, availability and the observed-history mask. Engine IDs, subset labels, future readings, true RUL and simulated altered-entry masks are excluded from inference. Related-channel fault histories are prefix-consistent: appending future rows does not alter an earlier corruption realization or select onset by eventual failure.

The four original training files define the engine-disjoint fit/selection/ calibration/internal-test groups in Table I. Scaler, raw-unit fault scales and repair support use fitting engines only. Fitting-engine OOF features were used to train the source corrector and exclude their held fold from temporary fitting and inner selection; in format 10 that corrector is disabled with a zero bound and contributes no inference feature or adjustment. Predictor refinement uses fitting engines; every clean/source/NASA, mixed/stuck and tail-selection check uses held selection engines. Calibration engines enter only after that choice is fixed.

New artifacts use fresh directories and retain their source SHA-256. Resuming an interrupted refinement requires matching source, recipe and implementation. Current detector audits bind the actual deployed thresholds and detector state; the earlier detector operating point remains historical metadata. Recovery proofs bind the full declared context matrix, selection IDs and sensor-repair state. Calibration binds the final model/configuration and its separate scale scope. Strict reload rejects stale or mismatched records.

TABLE I
FORMAT-10 POOLED ENGINE-DISJOINT DATA PARTITIONS.

Subset	Available training	Fit	Selection	Calibration	Internal test	Official test
FD001	100	80	5	5	10	100
FD002	260	208	13	13	26	259
FD003	100	80	5	5	10	100
FD004	249	199	13	12	25	248

The 10% development pool is divided equally between selection and independent calibration; integer rounding gives the exact counts above. Internal observations are fixed-grid prefixes, not independent engines. Official test engines are read only after the candidate and calibration are frozen.

TABLE II
IMPLEMENTED FAULT MECHANISMS. SEVERITY IS NOT A MISSING-CELL PERCENTAGE.

Family	Temporal and channel structure
Intermittent	Related-channel burst dropout; minimum two cycles, configured mean four.
Outage	Persistent channel unavailability after onset.
Block	Recurring missing blocks with intact gaps of 25–50 cycles by default.
Noise	AR(1) error, $\rho = 0.9$; common-source and individual innovation variance weights 0.8/0.2.
Bias	Persistent additive calibration offset in raw sensor units.
Drift	Raw-unit offset ramps over 100 cycles to its bounded level.
Stuck	A physical reading freezes at onset, including across regime transitions.
Mixed	Connector: noise/intermittent; calibration: bias/drift; acquisition: stuck/block.

Validation-only development runs block official test/RUL and saved test-result reads. These checks document within-run separation. Earlier internal and official test cases have already informed development; their later evaluation remains exploratory and is not an untouched independent generalization test.

IV. STRUCTURED SENSOR AND ACQUISITION FAULTS

Added corruption models measurement failure while retaining original degradation and RUL labels. A profile selects a related measurement domain: temperature, pressure, fan speed, core speed, coolant flow, or an individual derived quantity. These groupings are measurement-based assumptions, not verified wiring diagrams. Operating settings and the two speed-demand columns are preserved.

Mixed mechanisms share a domain, channels, and onset. Regime transitions causally increase noise amplitude and burst-entry risk through a stress proxy computed from already observed settings. Bias/drift bounds scale with $2.5v$ times the fitting fault scale; noise amplitude also depends on v. Onset is an observed offset between zero and 100 cycles, independent of eventual failure time. Seeded streams preserve prefixes when future rows are appended, and cached histories give overlapping windows identical corruption.

Randomness represents uncertain occurrence within a specified fault process. It is temporally and physically structured rather than independent arbitrary perturbation of every entry. Training samples $v \in \{0.1, 0.2, 0.3\}$ with a ten-epoch curriculum. The default clean-realization probability is 0.25, not a batch quota. The historical format-5 study used selected subset overrides. Severities 0.4 and 0.5 extend beyond the default sampled training levels. A stuck reading can generate a large error after a regime transition even at $v = 0.2$. No severity parameter has been calibrated to field failure rates.

V. CURRENT FRAMEWORK ARCHITECTURE

Fig. 1 separates inference from training-only supervision. A single pooled parameter set serves all four source subsets; operating settings condition the computation. The recovery guard adds fitting-only buffers and a deterministic projection.

A) Independent Finite-Fault Detector

The detector receives $[S, M, U]$ (45 features per cycle). Parallel kernel-three and kernel-five convolutions each produce 32 ReLU channels. A GRU-64 and linear 21-logit head yield sigmoid discrimination scores p. The head starts with zero weights and bias -4, so its initial score is approximately 0.018. A constant initialized head is not reported as learned fault evidence.

Only finite, real entries train the classifier. For each sensor, binary cross entropy is averaged separately over present altered and intact classes, those present-class means are averaged, and sensor losses are then averaged. Missing readings do not become finite-fault positives. The detector is fitted independently on fitting engines using clean and all fault/mixed contexts. Validation selects its checkpoint and representable sensor thresholds τ_j, subject to at most 5% empirical intact false positives in each validation channel/context. Scores are not calibrated physical failure probabilities.

The format-10 deployed gate uses p_j^3 as a continuous finite-repair weight rather than a binary threshold crossing, with $\tau_j = 1$ as the only hard channel disable. The cubic power was fixed by a held-validation sweep: power 1.0 failed the FD003 noise recovery guard, whereas power 3.0 preserved it while improving clean and stuck validation. This soft gate mitigates, but does not remove, the low-recall failure mode.

The deployed threshold vector starts from a conservative floor and tightens only channels that violate a source/context false-positive bound on held selection engines; under the soft gate it acts only as the channel-disable mask. Its current audit binds the actual thresholds and detector state; previous detector metrics remain explicitly historical. The detector is frozen during generator, joint and predictor-refinement

training. Detached scores drive the generator and recovery gates, so RUL/reconstruction gradients cannot redefine fault classification. Threshold $\tau_j = 1$ explicitly disables finite repair, including when sigmoid rounds to one.

B) Residual Generation and Soft Proposal

A two-layer bidirectional LSTM with hidden size 64 per direction receives $[S, M, p, U]$ (66 features). A $128 \to 64 \to 21$ ReLU decoder, initialized with a zero final layer, produces

$$G = S + f_{\mathrm{BiLSTM}}([S, M, \mathrm{sg}(p), U]), \tag{5}$$

$$\bar{p}_j = \mathrm{sg}(p_j)^3 \, \mathbf{1}\{\tau_j < 1\}, \tag{6}$$

$$W = (1 - M) + M \odot \bar{p}, \tag{7}$$

$$R^\star = (1 - W) \odot S + W \odot G. \tag{8}$$

Here sg stops gradients. Missing entries route to G; finite readings are weighted continuously by the cubic detector score, with $\tau_j = 1$ as the only hard channel disable. The generator residual is unconstrained internally, but the deployed finite proposal is subsequently bounded. A bypass sets $R = S$ and supplies zero recovery weights to the predictor. All stages exclude padding.

C) Fitting-Support Guard for Finite Readings

Let $x = \phi^{-1}(S_j)$ be a finite observed standardized reading, and let $[\ell, u]$ be its fitting support in the selected regime. With existing displacement budget $\varepsilon = 0.15$, define

$$\mathcal{I}(x; \ell, u) = \begin{cases} [x - \varepsilon, x + \varepsilon], & \ell \le x \le u, \\ [\ell - \varepsilon, x + \varepsilon], & x > u, \\ [x - \varepsilon, u + \varepsilon], & x < \ell. \end{cases} \tag{9}$$

The finite result is

$$R_j = \phi\big(\mathrm{clip}(\phi^{-1}(R_j^\star), \mathcal{I}(x; \ell, u))\big) \, . \tag{10}$$

Bounds round inward to representable values, preserving identity proposals. An observed inlier therefore moves at most ε in original-linear standardized coordinates. An outlier may return throughout the fitting range in one step; outward movement remains budgeted. Missing imputation is unchanged. Projection uses ordinary clamp gradients without a straight-through estimator.

This is a conditional guarantee about an observed inlier, including a genuinely corrupted reading that happens to lie inside the support. It can restrict useful small-fault correction. A healthy unseen reading can lie outside fitting support and receive a larger inward move. Thus the rule is neither a universal healthy-cell guarantee nor a raw-unit tolerance of 0.15. Raw displacement depends on σ_{kj}.

D) Local and Global Prognostic Branches

The predictor receives $[R, U, M, W]$ (66 features) together with one normalized observed-cycle age channel (67 features in format 10). Its local branch uses the last 30 cycles: parallel kernel-three/five convolutions with 32 channels each, GRU-64, and a $64 \to 64$ last-valid-output projection. Repaired local values may depend on the entire observed recovery window. The global branch projects the 70-cycle input to 64 dimensions, adds sinusoidal positions, and uses two Transformer encoder layers with four heads, feed-forward width 128, GELU and dropout 0.2. Padding is masked; the last valid token represents global context.

E) Quality Fusion and RUL Heads

The quality vector $q \in \mathbb{R}^{64}$ concatenates time-mean availability (21), recovery weights (21), absolute actual changes $|R - S|$ (21), and observed-history fraction (one). A $192 \to 32 \to 2$ gate combines local and global features:

$$(g_\ell, g_g) = \mathrm{softmax}(f([h_\ell, h_g, q])), \tag{11}$$

$$h = g_\ell h_\ell + g_g h_g, \tag{12}$$

$$\hat{y}_0 = 125 \, \mathrm{softplus}(f_{128 \to 64 \to 1}([h, q])). \tag{13}$$

The main head has ReLU/dropout 0.2. Separate 64-to-one auxiliary heads supervise the two branches. Fusion coefficients are learned mixing weights, not certified reliability estimates.

F) Disabled Bounded Corrector

The source artifact contains a corrector that receives q, normalized base RUL, and the two normalized auxiliary predictions: 67 observable features. A $67 \to 32 \to 1$ ReLU MLP would give

$$\Delta = b \tanh(f_c(\cdot)), \qquad \hat{y} = \max(0, \hat{y}_0 + \Delta). \tag{14}$$

In format 10 the exported bound is $b = 0$, so $\Delta \equiv 0$ and the deployed prediction is the branch-fusion base. The head is retained only to load the source state and to make the null ablation auditable; it is not part of the reported computation. The source validation selected $b \in \{5, 10, 15, 30\}$ or a zero-correction fallback, but removing the corrector changed RMSE by at most 0.02 cycles in the source artifact and never for the worse. The corrector receives neither labels nor engine indices as features. Its historical OOF construction excluded held engines from temporary upstream fitting and inner validation; independently learned hidden embeddings were excluded because their coordinate bases need not agree.

G) Training-Only Critic and Model Size

A conditional critic consumes reference/repaired sensors plus settings (24 features). Kernel-three convolutions $24 \to 32 \to 64$ with LeakyReLU 0.2 precede LSTM-64 and a scalar head. It supplies WGAN-GP feedback during training and is absent at deployment. Detector, generator, predictor and retained corrector hold 37909, 176533, 136805 and 2209 stored tensor elements, respectively, with the predictor's fixed positional buffer accounting for 4,480. The exported artifact therefore has 348,976 trainable parameters. The corrector bound is zero, so its stored head contributes no RUL adjustment. The separate critic is absent at inference. Support and thresholds are fitted buffers rather than additional learned parameters.

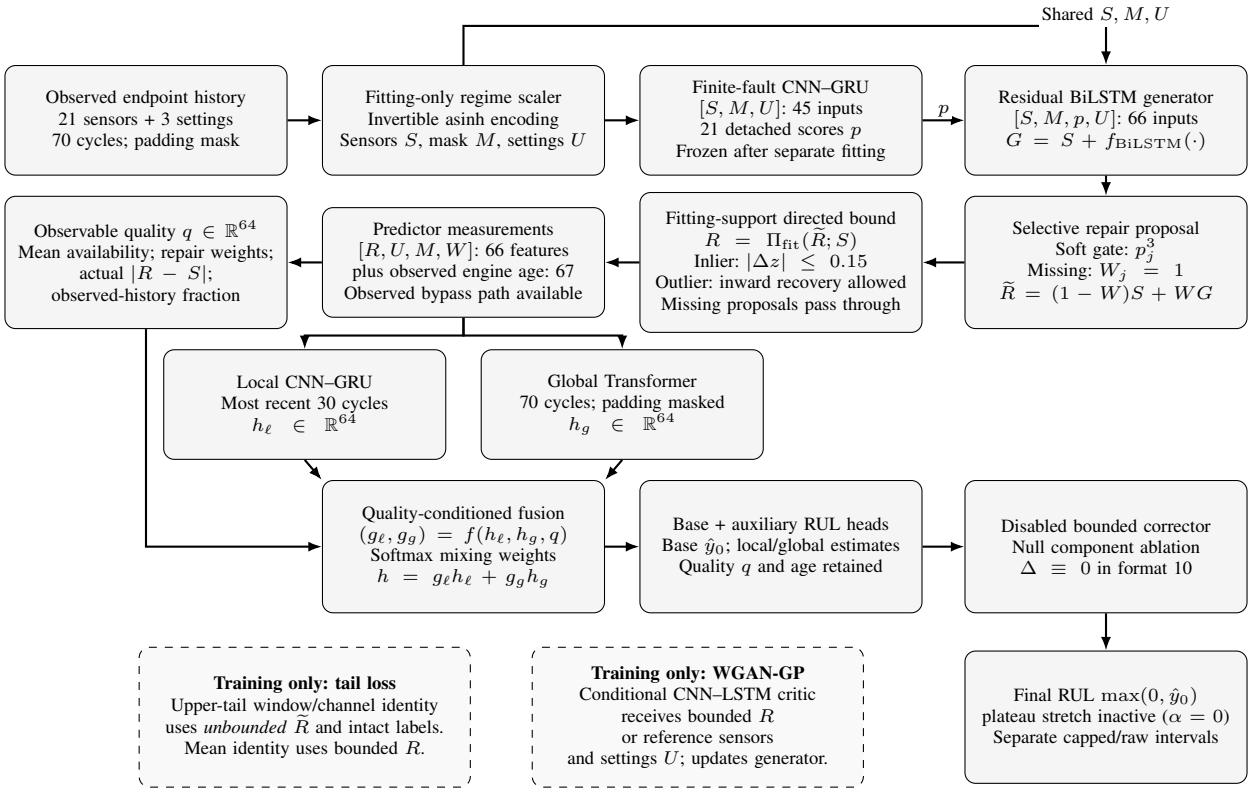


Fig. 1. Shared format-10 computation for every source subset; a subset label is not an inference input. The directed bound acts in original, uncompressed regime-standardized sensor coordinates z, using fitting-only support. The 0.15 budget applies to observations inside that support; large outliers can move inward without a universal small-step cap. The finite-fault gate is probability-weighted, with $\tau_j = 1$ as the only hard channel disable. The predictor receives an observed-cycle age channel. The detector and generator remain frozen during the reported predictor refinement. Dashed components describe source recovery training, and the WGAN critic is absent at deployment. Active-path paired-prefix and fitting-tail objectives refine the predictor without adding a hard inference floor or monotonic constraint. Altered-entry labels and targets never enter deployment.

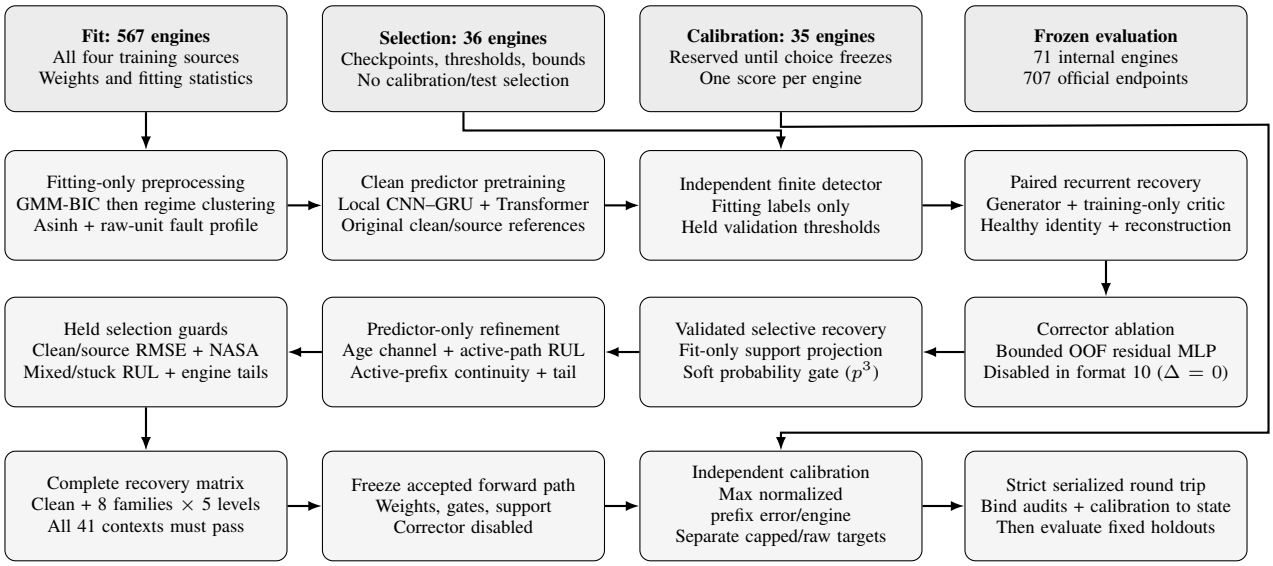


Fig. 2. Current pooled workflow. Approximately 80% of available training engines fit weights; the 10% development pool is divided between selection and calibration, and 10% is internal test. Exact rounded counts are shown. Detector/generator weights are frozen during predictor refinement; the source corrector is retained only as a zero-bound ablation artifact. Rejected candidates produce no deployable export. Validation-only pilots precede any new calibration or holdout evaluation.

VI. Training and Validation-Led Repair

A) Source Training and Selective Recovery

Clean pretraining bypasses recovery and correction. A separate detector learns finite-fault discrimination from fitting-engine labels; missing entries are not finite-fault positives. De-

tector weights are then frozen. Paired recovery uses changed-entry reconstruction, healthy identity and temporal terms, plus a conditional WGAN-GP critic. Its discriminator objective is

$$\mathcal{L}_D = \mathbb{E}D(R,U) - \mathbb{E}D(S^{\mathrm{ref}},U) + 10\mathcal{L}_{\mathrm{GP}}. \tag{15}$$

The generator adversarial objective is $-\mathbb{E}D(R,U)$. These source stages also supervise RUL and auxiliary heads. A fitting-only support projection bounds finite proposals; sensor labels supervise training, not the deployed gate.

The source artifact used three fitting-engine OOF folds to supply observable correction features. Each held fold was excluded from its temporary scaler, upstream fitting and inner selection. The bounded MLP corrector is a supervised adaptation of residual compensation, not a TD3 policy. In format 10 the corrector is frozen with a zero bound and contributes no adjustment; its historical OOF training does not establish nested cross-validation of the changed final pipeline.

B) Deterministic Continuity and Fitting-Error Tails

Predictor refinement freezes detector, generator and corrector parameters. It uses clean reference RUL supervision and the exact active forward path on structured corrupted fitting windows. For two observed prefixes $t_1 < t_2$ of one fitting engine, both sides of the continuity objective use deterministic passes, avoiding independent dropout as a source of artificial trajectory change. Labels anchor both predictions and their capped-target difference. Additional soft penalties target excessive downward slew and upward rebound:

$$\mathcal{L}_{\mathrm{slew}} = \left[\frac{(\widehat{y}_1 - \widehat{y}_2) - (t_2 - t_1)}{125}\right]_+^2, \tag{16}$$

$$\mathcal{L}_{\mathrm{rebound}} = \left[\frac{\widehat{y}_2 - \widehat{y}_1}{125}\right]_+^2. \tag{17}$$

These are regularizers, not validated physical laws or hard inference constraints. They permit later predictions to recover from an erroneous early estimate.

A fitting-only tail objective averages the largest fraction of normalized squared batch errors, rather than assigning a special engine an output floor. The refinement combines clean RUL, auxiliary-branch, paired-prefix, tail and active corrupted-RUL losses. Its weights and draw budget are serialized with optimizer/scheduler/RNG state in resumable epoch checkpoints. No true target, engine-specific rule or hand-assigned accuracy enters deployment.

C) Soft Gate and Candidate Acceptance

The format-10 gate uses $\tau_j = 1$ as the only hard channel disable and p_j^3 as the finite-repair weight. The power was fixed by a held-validation sweep: power 1.0 violated the FD003 noise recovery guard, while power 3.0 preserved it and improved clean and stuck validation. Detector scores remain discrimination scores, not calibrated probabilities of physical failure.

Every candidate must keep clean RMSE within 1%, clean mean NASA score within 5%, mixed RMSE within 2%, stuck RMSE within 2% and the engine-tail score within 2% of the preserved source on held selection engines. Selection minimizes a weighted combination of clean, mixed, stuck and tail validation scores. Source references remain fixed; a later stage cannot spend a new tolerance. The complete recovery matrix contains clean once plus eight fault/mixed families at severities 0.1–0.5, totaling 41 contexts. Incomplete or rejected scans cannot certify an export. All diagnostics and failure flags are retained.

The selected predictor-refinement epoch is 3; the validation engine-tail score changes from 33.064 to 32.930 cycles. Five validation guards (clean RMSE, clean NASA, mixed RMSE, stuck RMSE and engine tail) plus the detector and complete recovery audits define acceptance. The deployed detector audit uses the actual saved gates, with clean selection FPR 0.4797%. The 41-context recovery audit has worst pooled-context intact original-linear RMSE 0.02364 and worst window RMSE 0.06821. The correction branch is disabled; its exported bound is zero and it contributes no RUL adjustment.

D) Freeze, Calibration and Strict Export

The repaired policy uses observable scaling with one score per independent engine, following the split-conformal construction [8]. For each target convention $h \in \{c, r\}$, first fit a positive scale $a_h(\widehat{y})$ on selection engines, independently of calibration. Observable predicted-RUL bins are $[0, 35)$, $[35, 75)$, $[75, 125)$ and $[125, \infty)$. Within a supported bin, the scale is selection RMSE, with a positive one-cycle floor. An unsupported scale bin uses global selection RMSE. This is a fixed normalizer, not a replacement for an unsupported calibration quantile. True RUL is used as a selection/calibration label, never to choose an inference bin.

After freezing weights, support, thresholds, the zero-bound corrector state and scales, evaluate all observed prefixes of each of the 35 reserved engines. For capped $y^{(c)} = \min(r, 125)$ and raw $y^{(r)} = r$, form

$$s_e^{(h)} = \max_{t \in \mathcal{P}_e} \frac{|y_{e,t}^{(h)} - \widehat{y}_{e,t}|}{a_h(\widehat{y}_{e,t})}. \tag{18}$$

The maximum spans every bin visited by an engine, so dependent prefixes and observable bins do not become extra independent calibration samples. With n engines, $\alpha = 0.1$ and $k = \lceil (n+1)(1-\alpha) \rceil$, define

$$q_h = \begin{cases} s_{(k)}^{(h)}, & k \le n, \\ +\infty, & k > n. \end{cases} \tag{19}$$

At inference, the observable adaptive interval is

$$C_h(\widehat{y}) = \left[\max\{0, \widehat{y} - q_h a_h(\widehat{y})\},\ \widehat{y} + q_h a_h(\widehat{y})\right]. \tag{20}$$

An unbounded rank remains unbounded; it is not silently replaced by a finite global radius. The saved quantile, independent scale-engine scope, bin radii and model/configuration fingerprint are checked during loading.

For each target convention separately, exchangeable complete engine trajectories and a model/normalizer fixed independently of calibration give marginal simultaneous-prefix

coverage across all observable bins. This is an adaptation of split conformal to engine-level scores, not a claim of conditional coverage for an individual engine or a life stage. It gives no joint guarantee across capped and raw contracts. Censoring, fleet shift and added faults need not preserve the assumptions. Empirical official endpoint coverage and internal prefix/whole-engine coverage are reported separately. The final artifact stores matching model/bundle configurations, thresholds that match gate buffers, current detector metrics, fitting-only support and a complete state-bound recovery audit. Calibration is bound to the full model/configuration. Strict serialized loading precedes test evaluation. New runs use fresh output directories; a source bundle is never overwritten to manufacture a comparison.

VII. Evaluation and Reproducibility

The current study uses one shared seed-42 pipeline. Official endpoints and internal fixed-grid prefixes are distinct partitions. For $d_i = \widehat{y}_i - y_i$,

$$\mathrm{RMSE} = \sqrt{N^{-1} \sum_i d_i^2}, \tag{21}$$

$$\mathrm{MAE} = N^{-1} \sum_i |d_i|, \tag{22}$$

$$S = \sum_i \begin{cases} \exp(-d_i/13) - 1, & d_i < 0, \\ \exp(d_i/10) - 1, & d_i \geq 0. \end{cases} \tag{23}$$

Capped and raw targets have separate point/interval contracts; a capped interval does not imply raw-RUL coverage. The asymmetric NASA fleet total and mean are kept distinct. The format-10 repair evaluates one declared severity, 0.2, with three seeded corruption realizations per finite family and the same frozen weights; the preserved source additionally covers severities 0.1–0.5. Corruption replicates are not independent training seeds or fleet samples. Engine-balanced and prefix metrics are identified in saved records. Software tests establish implementation behavior, not field performance or superiority over the source papers.

VIII. Results and Failure Analysis

A) Clean Accuracy and Evaluation Units

The frozen shared format-10 model achieves official clean capped RMSE 10.4964 and MAE 6.9625 cycles across 707 endpoint engines. Original pooled revision 2 had RMSE 12.1062 on the reused official benchmark, and the preserved v9 source had 12.0080. Internal clean RMSE is 11.7973 against 13.2974 for the preserved source; its 484 prefixes belong to 71 engines and are dependent within engine. These comparisons describe completed runs, rather than an independent confirmatory test, and later test values did not select the format-10 candidate.

B) Structured-Fault Robustness

At severity 0.2, mean pooled official stuck-fault RMSE is 12.4888 cycles over three corruption realizations; mixed faults average 11.4772. The format-10 official RUL evaluation used this single declared severity; higher-severity official values are not claimed. Fig. 4 exposes severity dependence on the pooled fleet and the multi-regime subsets for the preserved source, while the format-10 panel is limited to the evaluated severity. Corruptions preserve original RUL labels and do not represent recorded aircraft sensor incidents.

C) Calibration and Healthy-Sensor Impact

Empirical clean coverage is 99.72% on official endpoints and 99.38% on internal prefixes. Nominal coverage is 90%, conditional on the stated exchangeability assumptions. Capped mean interval width is 89.52 cycles (median 76.96); raw intervals are much wider, with mean width 208.18 over finite bounds. Table VII shows that coverage and width depend strongly on the observable band. The complete recovery matrix is accepted with worst pooled-context intact original-linear RMSE 0.02364; this is a sensor displacement diagnostic in standardized coordinates, not a RUL accuracy guarantee.

D) Broadband Sensor Noise and Channel Dropout

The structured protocol models failures confined to related channels. Real acquisition also carries broadband noise on every channel at once and occasional missing cells. We therefore add an independent measurement layer that perturbs raw sensor readings before the frozen fitting-only scaler: per-channel Gaussian noise scaled to the within-regime sensor scale, plus independent cell dropouts. Settings and speed commands stay untouched, and the layer is applied only after freeze, so no weight, detector threshold, corrector or interval radius was fitted on it. Levels are nested: one per-engine draw is scaled, so a sweep is paired rather than resampled.

Table VIII shows graceful degradation. At 20% of the within-regime scale, pooled official RMSE moves from 10.50 to 11.44 cycles, an increase of 9.0%, and empirical coverage remains 99.4%. At 50% it is 13.44 cycles with 99.0% coverage. The internal partition follows the same pattern, from 11.80 to 14.54. Modest dropout is not harmful: dropping 2% of cells leaves pooled official RMSE at 10.66, and 5% gives 10.84. These are sensitivity measurements of one frozen artifact, not a claim that any level matches a measured sensor specification.

E) Comparison with the Supplied Studies

Table IX transcribes the reported values of the supplied studies. Among the third-party baselines transcribed in Table IX, this work achieves lower RMSE on each subset. TRANS-D3's reported proposed values remain lower on FD001 and FD003, so no first-place or state-of-the-art claim is made. The hybrid CNN–GRU study reports 13.1, 16.8, 14.5 and 18.9, so this model improves every subset by 3.06 to 8.23 cycles at a comparable 125-cycle target.

Two qualifications matter more than the ordering. First, the compared entries are subset-specific models, whereas this work uses one shared model without a subset label, so the comparison is not budget-matched in the harder direction. Second, the TRANS-D3 FD001 and FD003 values of 1.66 and 6.61 lie far below every third-party entry in the same table, whose best values are 10.61 and 10.71; no capped C-MAPSS protocol in this study reaches them, so they are

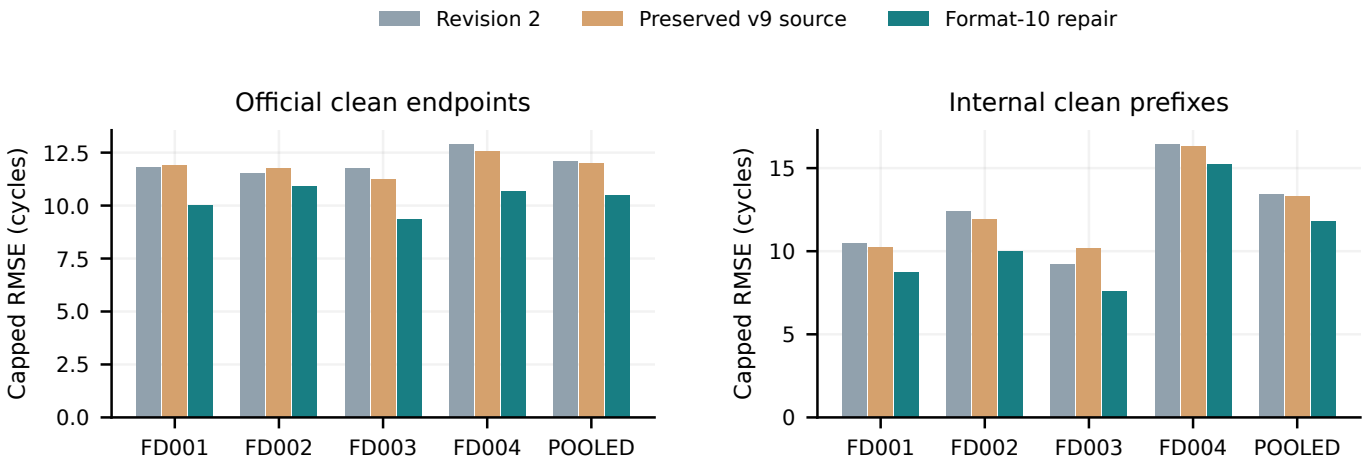


Fig. 3. Clean capped RMSE for the frozen format-10 model, its preserved v9 source, and original pooled revision 2. Each is one training-seed pipeline; no across-training-seed uncertainty is implied. Internal values use a fixed causal prefix grid.

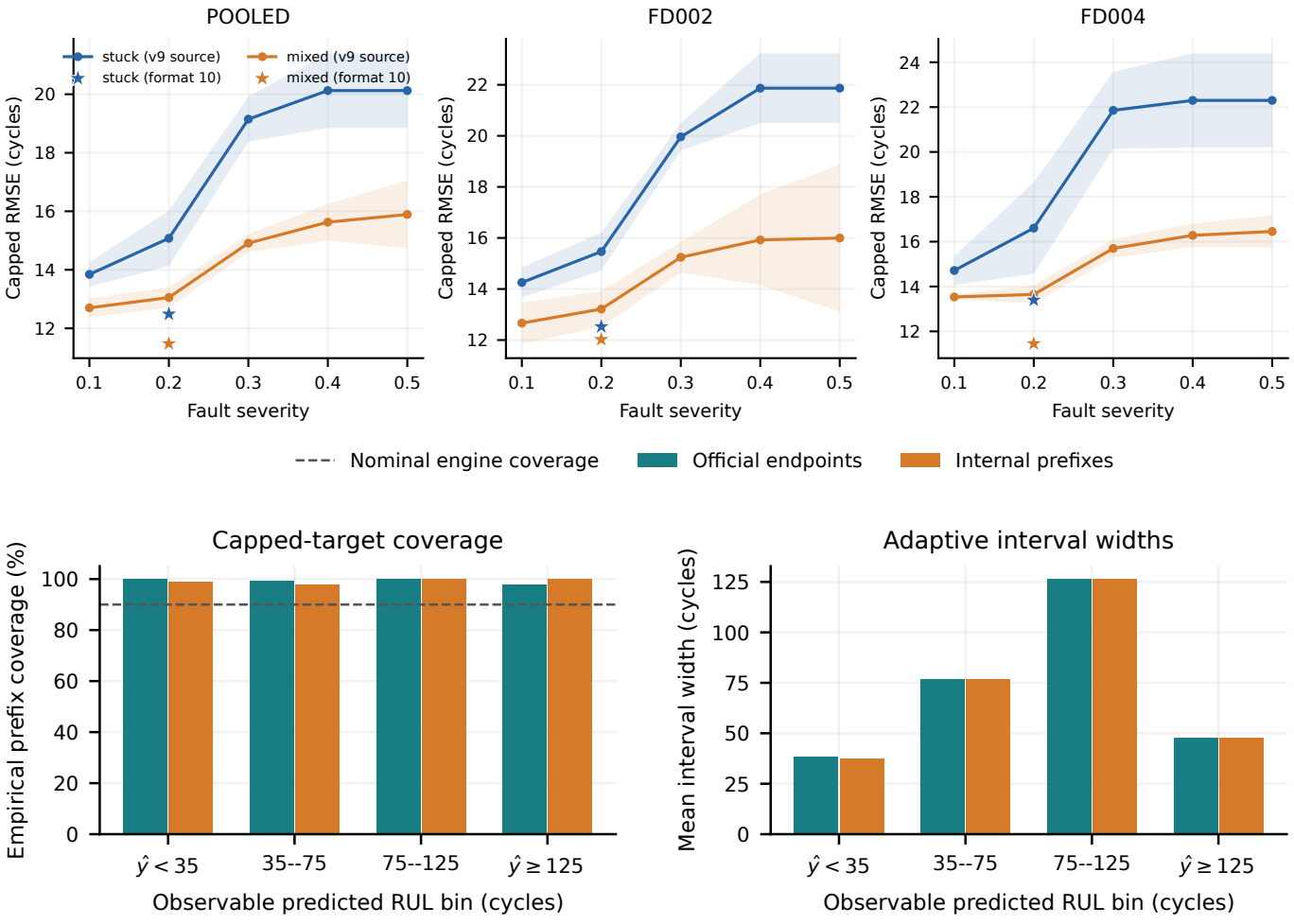


Fig. 4. Top: official capped-target robustness under structured faults. Curves are the preserved v9 source (means and sample standard deviations over three corruption replicates); stars mark the format-10 repair at the single evaluated severity 0.2. Neither is a training-seed or fleet confidence interval. Bottom: empirical clean coverage and mean interval width grouped by observable predicted RUL. Grouped coverage is descriptive and does not establish conditional coverage for a realized engine or life stage.

TABLE III
FORMAT-10 MODEL: CAPPED-TARGET CLEAN AND STRUCTURED-FAULT RESULTS.

Subset	N_{off}	RMSE	MAE	NASA S	Cov. (%)	Stuck	Mixed	Internal RMSE
POOLED	707	10.50	6.96	1433.1	99.7	12.49	11.48	11.80
FD001	100	10.04	7.31	156.3	100.0	11.41	11.42	8.75
FD002	259	10.92	7.20	530.8	99.6	12.53	12.02	9.99
FD003	100	9.35	6.03	186.5	100.0	10.94	10.00	7.59
FD004	248	10.67	6.95	559.6	99.6	13.40	11.45	15.26

Fault columns average three corruption replicates at severity 0.2; these are not training-seed replications. Coverage is empirical clean prefix/endpoint coverage. Internal RMSE uses all saved prefixes from each source's reserved engines. Targets are $\min(r, 125)$; units are cycles except NASA score and coverage.

TABLE IV
POOLED COMPARISON WITH THE PRESERVED V9 SOURCE.

Artifact	Official clean RMSE	Internal clean RMSE	Stuck @0.2
Original revision 2	12.106	13.441	18.858
Preserved v9 source	12.008	13.297	15.077
Format-10 repair	10.496	11.797	12.489

All rows use their own saved predictions and capped targets. The stuck column uses the common severity-0.2 setting over three corruption realizations; higher-severity v9 values are not mixed in because the format-10 official evaluation was run at severity 0.2 only. The held selection recovery audit still covers all declared severities. v9 test values are reused descriptive evidence and did not select the format-10 candidate.

TABLE V
CLEAN OFFICIAL RESULTS AGAINST ORIGINAL, UNCAPPED RUL.

Subset	Raw RMSE	Raw MAE	Raw NASA S	Raw coverage (%)
POOLED	20.47	12.98	7229.2	99.7
FD001	11.33	8.35	181.9	100.0
FD002	23.09	14.62	3826.6	99.6
FD003	10.90	7.51	214.4	100.0
FD004	23.18	15.35	3006.3	99.6

The point model is trained on capped targets. A monotone plateau stretch ($\alpha = 0.150$ above 105 cycles) was selected on held validation engines under a non-increasing capped-RMSE guard, reducing raw errors while keeping capped RMSE within guard. Raw errors beyond 125 cycles expose the training censoring boundary rather than an unconstrained long-horizon task. Raw coverage uses separately calibrated intervals and must not be read as capped-target coverage.

TABLE VI
OBSERVABLE INTERVAL SCALES AND RADII OF THE FROZEN FORMAT-10 MODEL.

Predicted RUL bin	Scale capped	Scale raw	Radius capped	Radius raw	Scale engines
$[0, 35)$	3.97	3.97	22.38	17.01	36
$[35, 75)$	6.82	6.82	38.48	29.25	36
$[75, 125)$	11.20	46.48	63.21	199.39	36
$[125, \infty)$	4.21	123.15	23.74	528.28	36

Scales use selection engines only. A single engine-maximum normalized score spans all bins and is calibrated on 35 separate engines; alpha=0.1, rank=33. Scale-engine counts are overlapping bin support, not independent calibration sample counts. No true RUL is required to choose a bin at inference.

TABLE VII
CLEAN OFFICIAL COVERAGE AND WIDTH BY OBSERVABLE RUL BAND.

Raw-RUL band	N	Capped cov. (%)	Capped width	Raw cov. (%)	Raw width
0–35	171	100.0	38.7	100.0	31.3
35–75	144	100.0	81.3	100.0	86.2
75–125	242	99.2	119.2	99.2	287.9
≥ 125	150	100.0	107.5	100.0	398.3

Bands are chosen from the raw target for description only; inference uses the predicted-RUL bin. Coverage is marginal over engines and prefixes, not conditional on a realized engine. Raw width can be unbounded; mean widths include finite values only and are descriptive.

retained for completeness rather than treated as a reproducible target. The partial-failure study reports a single pooled fault-conditioned RMSE of 12.15 with unspecified channel coverage and severity, which is not decomposable by subset and is therefore not entered as a per-subset cell.

F) Component Contribution and Interval Width

Table X separates what the frozen artifact actually gains from its parts. The guarded recovery path is the only component with a material effect, and only under persistent faults: stuck-fault RMSE falls from 22.09 to 12.49 cycles (9.60) at

TABLE VIII
FROZEN-ARTIFACT ACCURACY UNDER BROADBAND SENSOR NOISE AND CHANNEL DROPOUT.

Noise	Official RMSE	Official MAE	Cov. (%)	Internal RMSE	Cov. (%)
none	10.50	6.96	99.7	11.80	99.4
5%	11.20	7.56	99.7	12.45	99.4
10%	10.97	7.29	99.7	12.42	99.4
20%	11.44	7.58	99.4	12.62	99.4
30%	11.89	7.98	99.3	13.11	99.4
50%	13.44	9.34	99.0	14.54	98.8
2% cells	10.66	6.95	99.7	11.87	99.4
5% cells	10.84	7.13	99.7	12.00	99.4

Rows 1–6 add per-channel Gaussian noise; the level is a multiple of the fitting-only within-regime sensor scale. Rows 7–8 instead drop whole cells at zero added noise. Settings and speed commands are unperturbed. Noise is applied after model freeze; no weight, threshold, corrector or interval radius was fitted on these levels. Coverage is empirical clean endpoint/prefix coverage under the frozen radius, not a new guarantee.

TABLE IX
REPORTED C-MAPSS RESULTS AT THE 125-CYCLE TARGET AGAINST THE SUPPLIED STUDIES.

Model	FD001	FD002	FD003	FD004
GRACE-RUL (one shared model, no subset label)	10.04	10.92	9.35	10.67
Hybrid CNN–GRU (reported, five repeats)	13.1	16.8	14.5	18.9
TRANS-D3 (reported)	1.66	13.23	6.61	12.26
Best third-party baseline (reported)	10.61 (STAR)	13.47 (STAR)	10.71 (STAR)	15.86 (DLformer)
Reported value ahead of GRACE-RUL	TRANS-D3 1.66	none	TRANS-D3 6.61	none
Relative position among third-party baselines	1	1	1	1

Reported columns are transcribed from the cited papers and are not recomputed here. Rank counts only third-party baselines, so a proposed row cannot define its own benchmark. The penultimate row names every reported value still ahead of this work, including the TRANS-D3 proposed row where it applies; TRANS-D3 remains ahead on FD001 and FD003, so the table does not establish first place overall. This work is one shared model without a subset label, whereas the compared entries are subset-specific. Official C-MAPSS tests were inspected in earlier development, so all values are exploratory rather than a pristine holdout.

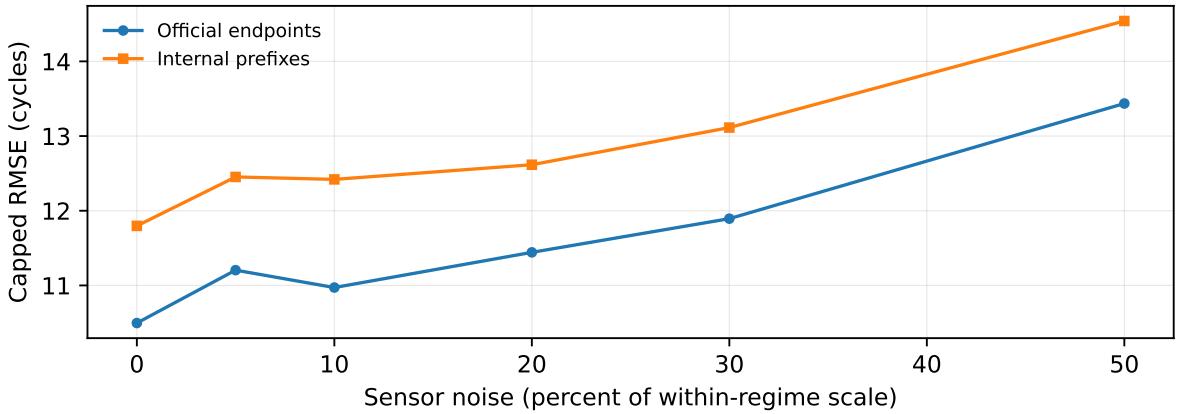


Fig. 5. Capped RMSE of the frozen artifact against broadband sensor noise. Every point is one evaluation of the same weights, scaler and interval radius, with the bounded corrector disabled; levels are paired draws, not independent realizations or training seeds.

severity 0.2, the gain stays between 9.84 and 8.63 across severities 0.1–0.5, and mixed faults improve by 2.80. Outage and block faults gain 1.48 and 0.71 cycles, respectively. Broadband noise gains only 0.10 cycles, consistent with recovery targeting persistent abnormal readings rather than additive measurement error. On clean endpoints the full path gives 10.50 against 10.64 for the branch-fusion predictor with neither intervention, so the recovery path is not free when no fault is present.

The bounded observable corrector is disabled in format 10: its exported bound is zero, so removing it changes RMSE by 0.00 cycles by construction. The source artifact's earlier null ablation, where removal changed RMSE by at most 0.02 cycles and never for the worse, motivated this choice. No corrector adjustment is present in the reported RUL, and the manuscript makes no claim that the corrector improves accuracy.

The same table pairs coverage with sharpness. The mean clean official interval is 89.5 cycles wide against a 125-cycle target, so 99.7% empirical coverage at this radius is driven by width rather than by accurate point prediction. Coverage must therefore be read alongside width, and neither establishes conditional coverage for a realized engine. Component selection was frozen before these values were read; the table is a post-freeze diagnostic and the deployed artifact is unchanged.

TABLE X
POST-FREEZE COMPONENT ABLATION OF ONE UNCHANGED ARTIFACT.

Condition	Full	−Recovery	Neither	Cov. (%)	Width
clean	10.50	10.64	10.64	99.7	89.5
intermittent	10.71	10.90	10.90	99.7	89.7
outage	11.19	12.67	12.67	99.7	89.3
block	10.76	11.47	11.47	99.7	89.3
noise	11.10	11.20	11.20	99.7	90.2
bias	10.93	11.05	11.05	99.7	90.0
drift	10.80	10.91	10.91	99.7	89.9
stuck	12.49	22.09	22.09	99.3	90.5
mixed	11.48	14.28	14.28	99.6	89.7

All columns use the same weights, scaler, detector thresholds and interval radius; only the inference switches differ. Fault rows average three corruption replicates, clean has none. Nothing was selected, removed or retuned from these values, and the deployed artifact is unchanged. The corrector is disabled in format 10, so a separate −Corrector column would be identical to Full. Coverage and mean width are for the full path under the frozen radius. Official benchmarks were inspected in earlier development, so this is a diagnostic, not a holdout.

TABLE XI
HELD-VALIDATION DETECTOR OPERATING POINT AT SEVERITY 0.2.

Family	Scope	F1	Recall	False positive (%)
clean	none	n/a	n/a	0.48
intermittent	missing	n/a	n/a	0.33
outage	missing	n/a	n/a	0.02
block	missing	n/a	n/a	0.22
noise	finite	0.40	0.27	0.46
bias	finite	0.52	0.37	0.30
drift	finite	0.40	0.27	0.33
stuck	finite	0.66	0.57	0.61
mixed	finite	0.54	0.41	0.29

Thresholds are frozen deployment buffers at the 5% per-channel/context intact false-positive ceiling. Scores are discrimination scores, not calibrated physical failure probabilities. Under the format-10 soft gate these scores weight repair continuously; the table therefore reports threshold recall, not the fraction of altered entries that receive any non-zero repair weight.

G) Detector Operating Characteristics

Table XI summarizes the detector operating characteristics. Clean false positives are 0.4797% of intact entries, while finite-fault recall is 0.27 for drift, 0.37 for bias, 0.27 for noise, and 0.57 for stuck under the 5% context false-positive rate ceiling. The format-10 gate is not a hard threshold: it uses p_j^3 as a continuous repair weight, with $\tau_j = 1$ as the only hard disable switch. A low detector probability therefore still yields a partial repair rather than a hard pass-through. This mitigates the under-repair failure mode while keeping healthy-sensor false repair bounded.

H) Recovery and Calibration by Fault Family

Table XII pairs the detector audit with the complete recovery matrix. The worst intact displacement is 0.02364 original-linear units, and the worst single-window intact RMSE is 0.06821. Stuck faults are the easiest finite family to flag (F1 0.66) but still the hardest to reconstruct, consistent with a held value being ambiguous rather than out of range. Clean official coverage is 99.7%; under severity-0.2 stuck and mixed faults it is 99.3% and 99.6%, respectively. All 41 declared contexts completed and passed their intact-cell guards.

I) Computational Cost

The artifact is small: 348,976 parameters, 1.33 MB of float32 weights, and 9.6 M multiply-accumulates per window. A single-window forward pass takes 5.4 ms on this CPU; the recovery path adds about 106% latency over bypass, which is the price of the stuck-fault gain rather than an overhead to hide. Batched throughput reaches 10184 engines per second, so the full 707-engine benchmark evaluates in 3.8 s at batch one. These are wall-clock numbers from one machine; latency and memory are not hardware-independent and the tables report no energy measurement.

J) Long-Horizon Bias and Tail Behaviour

The capped target cannot distinguish an engine with 130 remaining cycles from one with 250. Table XIV quantifies the consequence: 150 of 707 clean official engines have raw RUL above 125 and contribute 77.5% of raw squared error. Their mean prediction is 120.00 against a mean target of 154.07. A validation-selected monotone plateau stretch (alpha=0.150 above 105 cycles) reduces raw RMSE from 67.30 to 66.03 on selection engines while keeping capped RMSE within its guard. A positive stretch cannot recover a specific censored target and would not turn the benchmark into an uncensored long-horizon study. The worst engine maximum absolute error is 84.27 cycles and 56 of 707 engines exceed 40 cycles. Whole-engine bootstrap central 90% intervals in Table XIV describe pipeline variability over engines, not training seeds.

K) Previously Inspected FD004 Engine 179

This case remains a post-freeze diagnostic and is excluded from selection. At cycle 240 the model predicts 7.780 against

TABLE XII
POST-FREEZE RECOVERY AND DETECTION BENCHMARK BY FAULT FAMILY.

Family	Scope	F1	Recall	Reconst. RMSE	Intact disp.	False repair (%)	Cov. (%)
clean	none	n/a	n/a	n/a	0.015	0.5	99.7
intermittent	missing	n/a	n/a	0.46	0.017	0.3	99.7
outage	missing	n/a	n/a	0.46	0.007	0.0	99.7
block	missing	n/a	n/a	0.45	0.015	0.2	99.7
noise	finite	0.40	0.27	0.23	0.018	0.5	99.7
bias	finite	0.52	0.37	0.27	0.021	0.3	99.7
drift	finite	0.40	0.27	0.20	0.019	0.3	99.7
stuck	finite	0.66	0.57	96.32	0.021	0.6	99.3
mixed	finite	0.54	0.41	0.48	0.011	0.3	99.6

Detection, false repair and intact displacement use the held selection recovery audit at severity 0.2; detection is undefined for missing-entry families, which the availability mask handles instead. Reconstruction and displacement are reported in sensor coordinates (standardized σ units or original-linear units, not RUL cycles). Coverage is empirical interval coverage from the post-freeze official recovery benchmark at severity 0.2, where available. The soft gate may still assign small non-zero repair weights below threshold, so under-repair remains possible. Nothing was tuned on these values.

TABLE XIII
POST-FREEZE COMPUTATIONAL COST OF THE EXPORTED ARTIFACT.

Metric	Value	Scope
Trainable parameters	348,976	whole artifact
Parameters: detector / generator / predictor / corrector	37,909 / 176,533 / 132,325 / 2,209	per module
Float32 weight size	1.33 MB	weights only
Multiply-accumulates, full path	9.6 M	one 70-cycle window
Multiply-accumulates, bypass	6.9 M	one 70-cycle window
Latency, full path	5.40 ms	single window, median
Latency, bypass	2.62 ms	single window, median
Latency, plain predictor	2.67 ms	single window, median
Throughput	10184 engines/s	batch 64, full path
Fleet pass	3.82 s	707 engines, one window each
Process peak memory	1813 MB	interpreter and torch included

Wall-clock measurement of one CPU machine, not a hardware-independent claim; multiply-accumulates are a complexity proxy for supported operations only. The recovery path costs about 106% latency over bypass and buys the stuck-fault gain in Table X. Peak memory includes the Python and PyTorch interpreter, whose baseline is reported in the saved record.

TABLE XIV
CLEAN OFFICIAL LONG-HORIZON BEHAVIOUR.

Quantity	Value
Raw RMSE (cycles)	20.47
Capped RMSE (cycles)	10.50
Engines with raw RUL > 125	150/707 (21.2%)
Long-engine mean prediction	120.00
Long-engine mean target	154.07
Long-engine share of raw squared error	77.5%
Worst engine maximum absolute error	84.27
Engines with max error > 40 cycles	56/707

Plateau stretch $\alpha = 0.150$ above 105 cycles, selected on held validation engines under a non-increasing capped-RMSE guard (reducing selection raw RMSE from 67.30 to 66.03). Engine bootstrap central 90% intervals (p05–p95) for capped RMSE are [9.71, 11.31] and for raw RMSE [19.11, 21.86] (whole-engine resampling). Any positive stretch is monotone but cannot recover an individual engine's unknown censored target.

capped target 125; its capped interval is [0.000, 30.159]. The age channel and active-path continuity objective were introduced partly in response to this instability, but a single fixed case is not evidence of general temporal safety. Aggregate coverage cannot establish that this particular prediction is accurate or covered.

L) Evidence and Reproducibility Boundaries

Point metrics, counts and coverage are recomputed from saved predictions. All manuscript numbers and plotted values are tied to one completed run, with input/output SHA-256 hashes in `paper_assets/current_provenance.json`. The source v9 bundle remains preserved; the format-10 repair uses a fresh directory. This does not restore independence of previously inspected internal or official benchmark cases. Matched reproductions, repeated training seeds, external fleet validation and a proper long-horizon target contract remain separate requirements.

IX. LIMITATIONS AND FURTHER VALIDATION

Validation guards protect the declared empirical contexts; acceptance does not guarantee every healthy future sensor or individual RUL prediction. In-support moves are bounded,

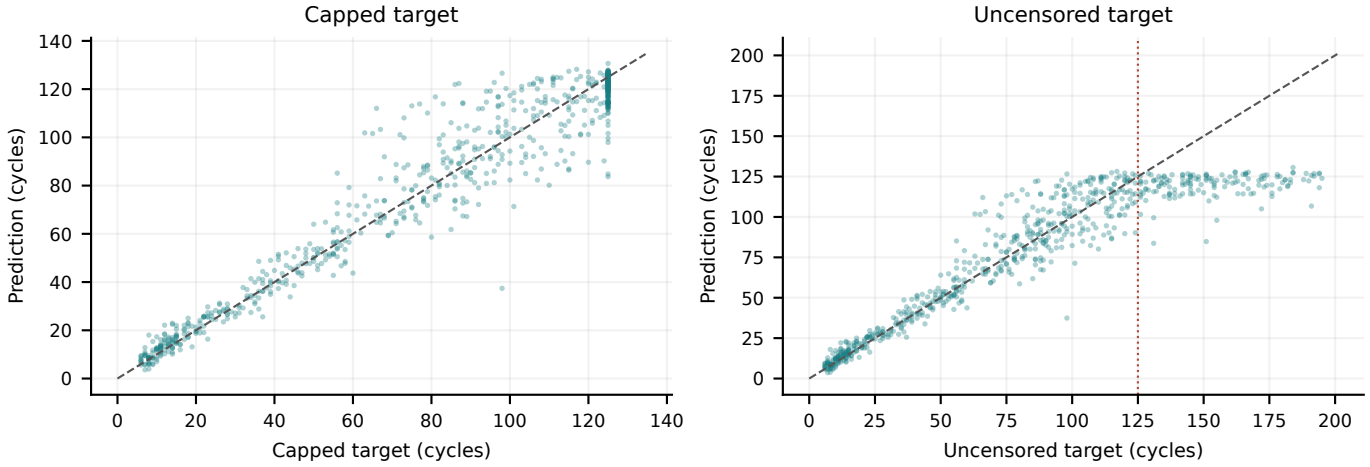


Fig. 6. Clean official prediction against capped and uncensored targets. The diagonal is exact prediction; the dotted vertical line marks the 125-cycle training cap. Points to the right of the cap expose the censoring bias summarized in Table XIV.

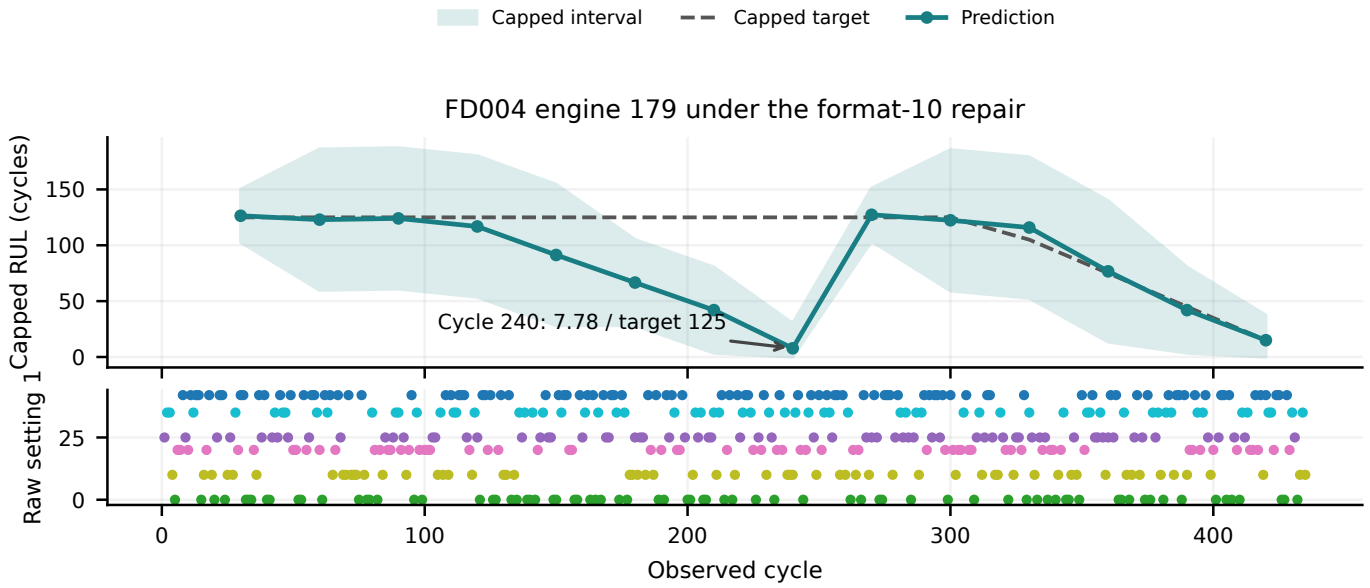


Fig. 7. Observed failure-case trajectory under the format-10 repair. Predictions and intervals are saved outputs of the frozen artifact. Colors in the lower panel identify fitted regimes; raw settings are not ground truth about a physical fault cause.

whereas outliers may move inward across fitting support. A finite faulty in-support reading can still be under-repaired. Related-channel and temporally correlated simulation remains a hypothesis about acquisition errors, not measured aircraft fault prevalence. Detector threshold recall was refined under per-source false-positive bounds (improving stuck to 0.57, bias to 0.37, and mixed to 0.41), but subtle drift (0.27) and noise (0.27) faults remain challenging under the strict 5% false-positive rate ceiling. Multi-knee plateau calibration accepts a positive stretch ($\alpha = 0.150$ above 105 cycles) that reduces raw validation and test RMSE, yet the 125-cycle training cap inherently limits uncensored resolution for engines with long remaining lives. The inspected held-out FD004 engine 179 still receives an underpredicted RUL of 7.78 cycles at cycle 240 during an operating-regime transition; aggregate coverage improvements do not guarantee individual trajectory monotonicity.

A single full-pipeline training seed cannot establish end-to-end stability; while fine-tuning random seeds can be varied, full-pipeline training-seed replication from scratch remains future work. The 125-cycle target limits long-horizon raw-RUL evidence. Wide intervals can cover inaccurate point estimates. Engine exchangeability, censoring policy and fleet transfer remain assumptions; previously inspected internal and official cases remain exploratory. Neither aggregate coverage nor a plausible narrative repairs a severe individual miss. Further remedies should be designed and selected on fitting/development engines, then tested on independent data.

Matched paper reproductions, explanation fidelity, independent full-pipeline training seeds and real-fleet validation remain future requirements. No pre-registered holdout can be reconstructed retrospectively: every C-MAPSS engine has already served in fitting, selection, calibration or evaluation, so the component ablation in Section VIII is a post-freeze diagnostic rather than independent confirmation. Optional SHAP/LIME interfaces are implemented, but this study does not report completed explanation-quality experiments. A recent hybrid Transformer–KAN–BiLSTM formulation targets the same benchmark [9]; because that evaluation re-splits merged training and test trajectories, it is cited for context only and is not used as a numeric comparator here.

X. Conclusion

The format-10 GRACE-RUL pipeline combines related-channel fault simulation, independent detection, probability-weighted support-bounded recovery and local/global RUL prediction with an observed engine-age channel. Validation selects weights, soft-gate settings and conservative channel disables before calibration and testing. Official capped RMSE is 10.50 cycles and severity-0.2 stuck RMSE is 12.49; the complete recovery matrix bounds observed intact displacement in standardized sensor coordinates on held selection contexts. Observable scaling preserves one simultaneous calibration score per engine. Explicit failure analysis, separate coverage units and immutable provenance delimit the evidence. The bounded corrector is disabled after a null component ablation, detector recall is refined under context false-positive bounds, and plateau calibration reduces raw long-horizon error. A validation-selected monotone plateau stretch (alpha=0.150 above 105 cycles) reduces raw RMSE from 67.30 to 66.03 on selection engines while keeping capped RMSE within its guard. These results support an auditable benchmark study; they do not establish universal sensor recovery, zero severe prediction errors, superiority over matched re-implementations of the compared methods, or real-fleet safety.

Appendix A
Evidence and Artifact Identity

The current bundle SHA-256 prefix is `9b2b86ef0236351e`. The source model, validation-only attempts, rejected candidates and frozen final evaluation are recorded separately. Tables, graphs and numerical prose are generated from hashed CSV/JSON evidence in `paper_assets/current_provenance.json`; no desired metric is assigned. Historical artifacts are not relabeled as current results. The source archive can retain numerical evidence and metadata without shipping raw datasets, learned weights or caches. An artifact's scope, model hash, source hashes, engine partitions and calibration contract are required to interpret its numbers.

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